MIL-F-43754B
31 March 1981
SUPERSEDING
MIL-F-43754A
7 MARCH 1974

MILITARY SPECIFICATION

FEED, HIGH CALORIC, MEDICATED, FOR MILITARY WORKING DOGS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the components and packing requirements for an extruded high caloric medicated dog food for use by the Department of Defense in feeding of the military working dog. The product may be held up to 9 months under normal non-refrigerated conditions of storage and distribution prior to consumption.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-P-704	-	Pails; Metal (Shipping, Steel, 1 through 12 Gallon)

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Non-Perishable Subsistence

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8710

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

FDA Bacteriological Analytical Manual, Bureau of Foods, Division of Microbiology, August 1978 5th Edition

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

PARR INSTRUMENT COMPANY

Technical Manual No. 130 - Oxygen Bomb Calorimetry and Combustion Methods

(Application for copies should be addressed to the Parr Instrument Co., 211 Fifty-third St., Moline, IL 61265.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

ASSOCIATION OF VITAMIN CHEMISTS

Methods of Vitamin Assay of the Association of Vitamin Chemists

(Application for copies should be addressed to the Interscience Publisher, Inc., a Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, NY 10016.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), the contractor shall furnish a sample for first article testing and approval (see 4.4, table IV and 6.2).

3.2 Materials. The product shall be prepared from ingredients which are clean, sound and free from mold, rodent filth and hair (hair other than that normally occuring in animal derived products), insect or insect part(s), paint, glass, metal, or any other evidence of contamination or off condition. Indgredients of vegetable origin shall be finely crushed before use. In addition, animal products, including dairy products, edible fats and oils shall be handled, processed and stored in accordance with good commercial practice so that flavor deterioration does not take place. The finished product shall contain at least 96 mg of diethylcarbamazine and 160 mg of styrylpyridium chloride per pound (454 g) of product.

3.3 Sanitary requirements. The product shall originate, be processed, marked and handled in plants which meet the requirements of MIL-STD-668, or in plants which are operated under the regulations governing the Animal and Plant Health Inspection Service, Agricultural Marketing Service, U.S. Department of Agriculture.

3.4 Processing. All materials shall be inspected to determine their wholesomeness and freedom from adulteration, contaminants and decomposition prior to blending. Plant inspection and processing procedures shall be under inspection supervision of the Military Veterinary Services or under inspection supervision approved by the Military Veterinary Services. The ingredients used and formulation shall be consistent with the end item requirements (see 6.3).

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3.5 Finished product. The product shall be uniform in color and texture, free from mold or recognizable spoilage and shall be free from foreign material such as, but not limited to dirt, stones, insects, insect parts, wood, paper, paint, glass, metal or charred particles. The pellets shall not be pasty or gummy in character and shall be 3/8 inch (10 mm) with a tolerance range of not less than 5/16 inch (8 mm) to not more than 9/16 inch (14 mm) in diameter.

3.5.1 Feeding tests. When ten or more healthy laboratory dogs (1 to 4 years old) weighing 70 to 90 pounds (32 to 41 kg) are fed the product in accordance with 4.6.2 the dogs shall sustain the average initial test weight so that the average terminal weight shall be at least equal to the average initial weight after 60 days. No individual healthy dog shall lose more than six percent of its starting weight. Each laboratory dog shall have been tested and show a negative Knotts test; negative fecal test for intestinal parasites; normal red, white, and differential blood cell counts; normal hemoglobin and blood urea nitrogen determinations; and a normal routine urinalysis.

3.5.2 Nitrogen balance and digestion coefficient. In accordance with the procedures specified in 4.6.3, four or more adult laboratory dogs shall maintain a nitrogen equilibrium or positive nitrogen balance when fed at a rate necessay to maintain body weight. This testing will only be conducted for first article qualification. In addition, minimum digestion coefficient percentage yield shall be as specified in table I:

TABLE I. Digestion coefficient percentage yield

	Not less than 1/
Dry matter	82
Protein	90
Fat	95
Nitrogen free extract (NFE)	80
Energy	87

1/ Results shall be reported to the nearest percent.

3.5.3 Storage capabilities (see 4.6.4 for procedures).

3.5.3.1 Fat stability. The iodine number shall exhibit a decrease not greater than 25 percent when tested in accordance with 4.6.4.1.

3.5.3.2 Grease-out or weeping. The product shall show no grease-out or weeping when tested in accordance with 4.6.4.2.

3.5.3.3 Mold growth. The product shall remain mold free when tested in accordance with 4.6.4.3.

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3.5.3.4 Insect and mite infestation. The product and container shall be free from viable eggs and larvae when tested in accordance with 4.6.4.4.

3.5.4 Caloric density. The product shall contain not less than 2200 digestible kilocalories per pound when tested in accoordance with 4.6.5.

3.5.5 Analytical requirements. The product shall conform to the analytical requirements of tables II, and III.

TABLE II. Chemical analysis

Based on 90 percent dry matter		
	Minimum	Maximum
Moisture 3/	6.0	11.0
Crude protein 3/	26.0	32.0
Crude fat 3/	25.0	30.0
Potassium	0.8	1.1
Ash 3/	4.0	6.5
Calcium 1/	0.85	1.4
Phosphorus 1/	0.7	1.2
Sodium	0.2	0.4
Crude fiber 3/	0.0	5.0
Nitrogen free extract (NFE) 2/	15.50	39.00

1/ Minimum ratio 1:1 (calcium to phosphorus).

2/ NFE is that portion of product remaining after the water, protein, fat, ash and fiber have been removed. NFE is interchangeable with carbohydrates.

3/ Label guarantees, within range of maximum and minimum, see 5.4.

TABLE III. Minimum amounts per pound of product

Based on 90 percent dry matter	
Minerals	Minimum
Magnesium	454.0 (mg)
Iron	50.0 (mg)
Manganese	4.5 (mg)
Zinc	20.0 (mg)
Copper	5.0 (mg)

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TABLE III. Minimum amounts per pound of product (cont'd)

Based on 90 percent dry matter	
Vitamins	
Vitamin A	4150 (International Units)
Vitamin D	400 (International Units)
Vitamin E	64.0 (mg)
Thiamine	0.8 (mg)
Riboflavin	1.9 (mg)
Niacin	9.6 (mg)
Pyridoxine	1.0 (mg)
Pantothenic acid	8.0 (mg)
Folic acid	0.2 (mg)
Choline	960.0 (mg)
Vitamin B_{12}	0.032 (mg)

3.5.6 Microbiological requirements. The product shall comply with the following:

Aerobic plate count (APC): Three of five samples shall be equal to or less than 100,000/gm and 2 of 5 samples shall be no greater than 1 x 10^6/gm.
Coliforms: Five of 5 samples shall be no greater than 120/gm. Most Probable Number (MPN).

3.6 Storage of manufactured feed. The product shall be packaged and packed promptly after processing and shall not be stored more than 90 days prior to shipment.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification. The product shall be unacceptable if not produced and stored in plants which meet the requirements specified in 3.3.

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be tested for the characteristics specified in table IV. The failure of any test shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

TABLE IV. First article tests

Characteristic	Requirement paragraph	Test method	Number of determinations	Results reported as
Feeding test	3.5.1	4.6.2 and 4.6.2.1	See requirement	Pass or fail
Nitrogen balance and digestion coefficient	3.5.2	4.6.3	See requirement	Pass or fail
Iodine value (fat stability)	3.5.3.1	4.6.4.1	2	Pass or fail
Grease-out or weeping	3.5.3.2	4.6.4.2	1	Pass or fail
Mold growth	3.5.3.3	4.6.4.3	1	Pass or fail
Mites or insects	3.5.3.4	4.6.4.4	1	Pass or fail
Caloric density	3.5.4	4.6.5	2	Pass or fail
Analytical requirements	3.5.5 Tables II and III	4.6.6	2	Pass or fail

4.5.1 Componenet and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.5.1.1 Examination of components. Each ingredient shall be examined organoleptically as necessary, to determine conformance to the condition requirements of 3.2. The sample unit for examination shall be an approximate 1-pound (0.45 kg) composite of each major ingredient (constituting at least 1 percent of the formulation) and a 4-ounce (112 g) composite of each minor ingredient (under 1 percent of formulation) derived from five primary containers. In the event a lot consists of less than five containers, the sample unit shall be composited from each container in the lot. Nonconformance of any ingredient to the applicable requirements of 3.2 shall be cause for rejection of the component lot or the involved lot of finished product.

4.5.2 In-process examination. Examination shall be made during processing to determine compliance with the applicable requirements of 3.2 and 3.4. Records of all applicable control data shall be maintained. Noncompliance with one or more of the above referenced requirements reflected by actual examination or by examination of records shall be cause for rejection of the involved lot of finished product.

4.5.3 End item and packaging inspection. Examination of the end item shall be in accordance with tables VI through IX, and 4.5.3.1 through 4.5.3.4. The examination criteria for the above mentioned tables shall be as shown in table V. Acceptable quality levels (AQLs) shall be expressed in terms of percent defective for tables VI and VIII and as defects per hundred units for table VII.

TABLE V. End item and packaging examination criteria

Table	Inspection level	Sample unit	Lot size expressed in in	AQLs		
				Major A	Major B	Minor
VI	S-3	1 shipping container (pail)	Shipping containers	-	-	2.5
VII	S-2	Contents of 1 shipping container (pail)	Shipping containers	1/	1.5	-
VIII	S-3	1 filled and sealed pail	Pails	0.65	-	-

1/ Finding of one or more major A defects shall be cause for rejection of the lot.

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TAABLE VI. Examination for net weight 1/

Category	Defect
Minor	
201	Less than 24 lbs. 12 ounces (11.23 kg). 2/

1/ Lot shall be rejected if sample data indicate lot average weight is less than specified net weight.

2/ Report to the nearest 1 ounce (0.01 kg).

TABLE VII. Examination of product characteristics

Category		Defect
Major A	Major B	
101		Presence of foreign material, e.g., dirt, stones, insect(s), insect part(s), wood, paper, paint, glass, metal, charred particles etc.
	151	Not uniform in appearance as concerns color and texture.
102		Evidence of mold or spoilage, i.e., a typical odor, color, etc.
	152	Pasty or gummy.

TABLE VIII. Examination of leakage 1/

Category	Defect
Major A	
101	Evidence of leakage.

1/ See 4.6.1 for procedure.

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4.5.3.1 Examination of pellet diameter (size). Examination shall be made of pellet diameter to determine compliance with the requirements of 3.5. Measurement shall be on a go, no-go basis using a suitable constructed gauge. The sample unit shall be a 5-pound (2.27 kg) composite of product derived from intermittent and approximately equal sampling periods during a day's production. Fifty pellets shall be taken from the composite sample and measured for size. Four or more pellets with diameters not within the specified tolerances shall be cause for rejection of the lot.

4.5.3.2 Examination of storage time prior to shipment. Examination shall be made of storage time to determine compliance with the requirements of 3.6. Records of storage time shall be maintained. Nonconformance indicated by examination of records or date of pack, or absence of pertinent information shall be cause for rejection of the involved lot.

4.5.3.3 Examination of shipping containers (pails). Examination of the filled and closed shipping containers shall be in accordance with the applicable examination criteria of PPP-P-704. In addition, the following defects shall be included in the table of examination: Major: (1) National stock number, item description, contract number or date of pack markings missing, incorrect or illegible; (2) Plastic bag missing or not as specified (when applicable); (3) Alternative metal pail not constructed or interior coated as specified (compliance ascertained by review of labels, invoices or similar valid documents). Minor: (1) Other required markings missing, incorrect or illegible; (2) Plastic bag thickness less than specified. Commercial packs shall be examined only for the marking defects specified above.

4.5.3.4 Examination of unit loads. Unitized loads shall be examined in accordance with the criteria of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

4.5.4 End item testing. The finished product shall be tested for conformance to the characteristics shown in table IX. The sample for testing analytical requirements shall be a composite made up of individual approximate 1-pound portions from the number of primary containers indicated by inspection level S-2. The lot size shall be expressed in terms of primary containers. A lot shall not exceed one day of production. The samples for testing microbiological requirements shall be a 1-pound sample obtained aseptically from 5 primary containers and placed in separate impervious sterile sample containers. Each sample container shall be tested for aerobic plate count and coliform (using the 3 tube MPN technique). Results shall be reported to the same unit or decimal as specified for the pertinent requirement, except that for Vitamins A and D shall be reported to the nearest 50 International Units. All microbiological results shall be reported in accordance with referenced procedures specified in 4.6.7. Nonconformance to one or more of the microbiological (see 3.5.6) and analytical requirements (see 3.5.5) shall be cause for rejection of the involved lot of finished product.

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TABLE IX. End item tests

Characteristic	Requirement paragraph	Test method
Analytical requirements	Table II	4.6.6
Microbiological requirements	3.5.6	4.6.7

4.6 Methods of inspection.

4.6.1 Leakage examination. The filled and closed pail shall be submerged in water (100 ± 5°F (38 ± 3°C) initially rotated end over end three times to free entrapped air then positioned so that the pail is lying on its side with not less than 3 inches (76 mm) of water over any portion of the pail, and not less than 1 inch (25 mm) of water between the lower portion of the pail and the bottom of the container. The pail shall be held in this position for not less than 30 seconds and shall then be rotated so that observation for leaks cna be made over the entire circumference of the pail. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air shall not be considered as a sign of leaks.

4.6.2 Feeding test. The feeding tests shall be conducted at a location designated by the producer. All test dogs shall receive a physical examination by a doctor of veterinary medicine within thirty days prior to the test period (see 3.5.1). The kennel facility should provide an area of isolation confinement for each dog during part of each day. Physical examination shall include the following and shall be recorded.

a. Age of dog.
b. Condition of dog (to include actual weight, recommended weight, texture and condition of hair coat and skin).
c. Knotts test.
d. Fecal test.
e. Total and differential red and white blood cell counts.
f. Hemoglobin determination.
g. Blood urea nitrogen.
h. Urinalysis.

In addition, a general summarization of kennel facilities shall be recorded and shall include:

a. Condition and state of repair.
b. Location in relation to high sound levels.
c. Description of design (is each dog in isolation confinement? is ventilation sufficient to provide comfort?).
d. Diet (indicate what diet or feed the dog has had during the 30 days prior to the test).
e. Weather data complied daily from 30 days prior to the test and continued throughout the testing period to include:
   (a) The daily high and low temperatures.
   (b) The daily high and low humidities.

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4.6.2.1 Feeding test procedures. The test procedures shall be as follows:

a. Dogs are to be fed the test food ad libitum for a period of seven days.
b. Potable drinking water shall be provided separately or on an ad libitum basis.
c. Weight of the dogs shall be recorded to the nearest pound (kg) and the feed shall be weighed to the nearest ounce (gram).
d. The dogs are to be weighed at the end of the seven days ad libitum feeding period. This will be considered as the initial test weight.
e. Determine the average initial test weight of the group.
f. The average quantity of food fed each dog will be from 1.0 to 1.25 pounds (454 to 567 g) per day.
g. For a period of 60 days thereafter, the dogs must consume the test food at an average of not less than 30 K calories per pound of dog per day.
h. In comparing the average initial and average terminal test weights, the average initial weight will be sustained. The average terminal weight will be at least equal to the average initial weight. No individual dog will lose more than 6 percent of its initial weight.
i. All test and weight records including observations regarding character of the stools shall be maintained and made available for inspection.

4.6.3 Nitrogen balance and digestion coefficient percentage. Minimum digestion coefficient percentage requirements of 3.5.2 may be obtained during the nitrogen balance study to satisfy the specified requirements. The following method shall be used: Not less than four adult laboratory test dogs shall be housed individually in raised, screen-bottomed dog metabolism cages and standardized by feeding the test food for not less than fifteen days at a rate of 37 K calories per pound body weight per day or not more than 1.25 pounds (0.57 kg) for an 80 pound (36 kg) dog, or equivalent. On the fifteenth day, a fecal marker, such as carmine or charcoal shall be mixed into the diet (for that day only) in sufficient amounts to enable separation of feces containing the marker from non-marked feces. Except for unmarked feces appearing on day 16, total collecion of feces and urine shall be accomplished for the ensuing seven days; on a daily basis. On the 22nd and 23rd days of total feeding the marker shall again be mixed into the diet as before. Marked feces appearing subsequent to the 22nd day's feeding shall be discarded. Urine collection shall begin with the first meal fed during the collection period and shall be terminated immediately prior to the first meal subsequent to the collection period. All collections shall be made quantitatively. The urine shall be collected under toluene and stored in the refrigerator until the end of the colleciton period. The feces shall be stored frozen in a closed container until analysis. At the end of the collection period, the daily collecions of urine shall be combined for each dog and total volume recorded. The feces shall be thawed, weighed and homogenized. It may be advantageous to add an equal weight of distilled water to the feces to assist in homogenization and subsequent sampling. The urine shall be analyzed for total nitrogen. The feces shall be analyzed for total nitrogen, dry matter, fat, ash, and fiber. The food shall be analyzed for dry matter, total nitrogen, fat, ash, and crude fiber. Gross energy shall be determined by bomb calorimetry. Nitrogen-free extract shall be calculated by

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difference of 100 - (percent dry matter + percent protein + percent fat + percent ash + percent fiber). The calculation of digestion coefficient shall be determined by substracting the amount of the nutrient found in the feces from the amount fed, divided by the amount fed and converting to percentage. In the case of available energy, the caloric content of the feces as determined by bomb calorimetry shall be subtracted from the gross energy consumed based upon the bomb calorimeter value of the food and divided by the gross energy intake and converted into percent. The dog will be adjudged to have been maintained in nitrogen equilibrium or in positive nitrogen balance if the sum of nitrogen excreted in the urine and in the feces did not exceed the food nitrogen intake during the period. Nitrogen balance shall be stated in grams of nitrogen lost or gained.

4.6.4 Storage capabilities.

4.6.4.1 Fat stability. A 1-pound (0.45 kg) sample shall be incubated at 120°F (49°C) for 120 days. The iodine number shall be determined on a 0.8-g sample of the product in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists.

4.6.4.2 Grease-out or weeping. A 15-pound (6.8 kg) sample shall be incubated for 120 days at 120°F (49°C). The product shall be tested visually for compliance with this requirement.

4.6.4.3 Mold growth. A 15-pound (6.8 kg) sample shall be incubated for 120 days at 80°F (27°C) and 85 percent relative humidity. The product shall be tested visually for this requirement.

4.6.4.4 Insects and mites. A 15-pound (6.8 kg) composite sample of the product shall be incubated for 120 days at 80°F (27°C) and 85 percent relative humidity. After incubation, the contents of a 1-pound (0.45 kg) sample shall be sifted through a 12-mesh sieve and the material collected in shallow tray. The sifted material shall be spread evenly on a sheet of wax paper and examined under 20X magnification.

4.6.5 Caloric density. Caloric density shall be determined by bomb calorimetry.

4.6.6. Analytical requirement tests. Chemical analysis shall be made in accordance with the following methods in Official Methods of Analysis of the Association of Analytical Chemists. Analysis for vitamins shall be made in accordance with Methods of Vitamin Assay of the Association of Vitamin Chemists and gross energy shall be determined in accordance with Manual No. 130, Oxygen Bomb Calorimetry and Combustion Methods.

Test	Method
Moisture 1/	Vacuum oven
Crude protein 1/	Kjeldahl
Fat 1/	Ether extraction
Fiber 1/	Crude fiber
Ash 1/	Ash

1/ Chapter; Animal Feeds.

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Mineral	Method
Calcium	Atomic Absorption (Spectrophotometry)
Magnesium	"
Iron	"
Copper	"
Manganese	"
Cobalt	"
Zinc	"
Iodine	Iodine in mineral mixed feed
Phosphorus	Phosphorus in feed
Potassium	Flame Photometric under Fertilizer
Sodium	" " "

Vitamin	Method
Vitamin A	Colorimetric
Carotene	Chromatrographic
Thiamine	Thiochrome
Riboflavin	Fluorometric
Choline	Reineckate
Vitamin D	Chemical
Vitamin E	Chemical
Niacin	Microbiological
Pantothenic Acid	"
Pyridoxine	"
Folic Acid	"
Vitamin B_{12}	"

4.6.7 Microbiological test. Microbiological testing of the product shall be made in accordance with the current edition of FDA Bacteriological Analytical Manul Bureau of Foods, Division of Microbiology.

5. PACKAGING

5.1 Preservation. Preservation is not required.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.1).

5.2.1 Level A. A net weight of 25 pounds (11.3 kg) of product shall be unit packed in a 5-gallon capacity metal pail conforming to type II or III of PPP-P-704. A net weight tolerance of minus 4 ounces (113 g) is permitted, provided the average net weight of pails in any lot is not less than 25 pounds (11.3 kg). Prior to filling, each pail shall be lined with a minimum 0.005-inch (0.13 mm) thick plastic bag fabricated of material conforming to class 2 of L-P-378. The pail liner shall be securely closed after filling. Alternatively, when the interior of the pail is coated overall with a food grade coating suitable for the product, the plastic bag liner will not be required. The pail shall have an exterior coating of olive drab color and varnish. Each pail shall be closed in accordance with the appendix of PPP-P-704 and

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shall not leak when examined in accordance with 4.6.1. Alternatively, the 5-gallon (19 L) capacity metal pail shall be fabricated of 0.25 pound per base box electrolytic tinplate (5.6 g/m^2). The pail shall have a plate base weight of 118 pounds and be fitted with a bail and handgrip. The cover shall have 16 lugs and have a gasket flowed into the underside of the outside edge of the removable cover so when closed the pail shall not leak when examined in accordance with 4.6.1. The interior of the pail shall be coated overall with a food grade coating suitable for the product. The alternative pail shall have an exterior coating of olive drab color and varnish.

5.2.2 Commercial. The product shall be unit packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in acordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling. Each unit pack shall have labeling that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. The label shall include guarantees for crude protein, crude fat, crude fiber, ash and moisture in accordance with table II, footnote 3. In addition, each unit pack shall be labeled or marked with the following feeding directions:

FEEDING DIRECTIONS

Maximum Stress Diet is a concentrated food. The quantity required to maintain body weight is considerably less than other dog foods and will vary for each dog depending on environment, activity, temperament and stress factors. AVOID OVERFEEDING.

Feed Sufficient Quantity to Maintain Ideal Body Weight

(Most 70-pound dogs require 14-20 ounces (0.04-0.57 kg) daily)

To determine average initial feeding:

Maximum Stressed Dogs	$\frac{\text{Body weight in pounds}}{4}$	= ounces to feed daily
Medium Stressed Dogs	Divide body weight by 4.5	
Low Stressed Dogs	Divide body weight by 5	

Supplemental diethylcarbamazine and styrylpyridium chloride is not required unless supplemental, nonmedicated food is fed.

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5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1).
(c) Applicable level of packing (see 5.2).
(d) The type and class of unit load when unit loading is required (see 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.3 Alteration of formula. Formulation will be established with the production batch when first article testing of this batch is satisfactory. Alterations in the ingredients of the formulation may be made only with the approval of the contracting officer, and the substituted ingredients are of an equal or higher quality. Slight alteration in the amounts of ingredients is permissible to compensate for variations in nutrient content of ingredients.

6.4 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 50

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 8710-0020

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-43754B Feed, High Caloric, Medicated, For Military Working Dogs

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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C. REASON FOR RECOMMENDED CHANGE(S)

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